Replication of weight-reading experiments 1 and 2
November 2, 2026
This report and its interpretation were prepared by the OpenAI assistant (Codex). They are not user-authored and do not imply the user's endorsement. Model responses are linked separately and preserved unchanged.

What was reproduced

Experiments 1 and 2 from Sparks of Weight-Space Programming were repeated using the seven released prompts: four policy networks for game identification and three diffusion networks for shape identification. The source release was pinned at commit 4d09ba6d.

The tested models were GPT-6 Astra, Claude Opus 5.5, and Claude Fable 5.1. Their API identifiers and complete settings are preserved in the transcripts. They were accessed through the official OpenAI Responses and Anthropic Messages APIs. Each request contained the original user prompt, including all numerical weights, and the neutral system instruction "Answer the user's request." Prompt hashes were checked against the archived source manifest. Each trial was a fresh request, with no tools, web access, code execution, demonstrations, or feedback. Ground-truth labels and previous answers were not included in the requests.

Astra used high reasoning effort with detailed reasoning summaries. The Claude models used adaptive thinking with high effort. These settings do not establish equal inference compute. No temperature or generation seed was specified.

This is a replication of the released weight-reading tasks and category outcomes. The underlying networks were not retrained. Experiments 3–8, training procedures, and the authors' historical Codex CLI execution environment were not reproduced.

Results

Model	Games	Shapes	Filled descriptions
GPT-6 Astra	4/4	3/3	0/3
Claude Opus 5.5	4/4	3/3	2/3
Claude Fable 5.1	4/4	3/3	1/3

The primary results reproduce the reported category-identification success on all seven released inputs for Astra and extend it to both Claude

models. Game labels treat Freeway and Frogger as equivalent. Shape scoring accepts the published category aliases, including square/rectangle.

Fable's Pong answer was manually adjudicated as correct: it explicitly identifies Pong-style, single-player Pong and describes the vertical paddle, but also calls it "Breakout-without-bricks" as an analogy. The automatic scorer flags both names. This assistant review was not blinded to the model or answer key; its rationale is retained in the linked scoring records. All other final category results were scored automatically. Unicode hyphens were normalized for label matching.

All three diffusion targets generate filled shapes. Astra describes all three as outlines. Opus describes the square and triangle as filled, but the circle as a ring. Fable describes the triangle as filled and the other two as outlines. The last table column measures only this attribute of the written description; it is not a complete image-fidelity score. Dimensions, variability, drawings, and individual numerical claims have not been independently scored.

Execution qualifications

The initial pilot had an assistant-imposed output limit of 16,384 tokens. Opus's Breakout response identified the correct game but was truncated during its description. At the user's direction, the smaller limit was removed. Subsequent OpenAI requests omitted `max_output_tokens`; Anthropic's required `max_tokens` was set to its documented maximum of 128,000. The truncated Opus trial was rerun and completed with the correct answer, using 18,161 output tokens including thinking. Earlier completed responses were retained. Thus output limits were not uniform across trials, and this was not a fully preregistered comparison. Both Opus attempts are preserved.

The first Astra request was lost because the runner incorrectly treated an `error: null` field as a failure. Its task outcome and exact usage are unknown; it is excluded from scoring and cannot be supplied as a transcript. A separate failed local network launch is also documented. Neither is treated as evidence about model capability. There are 21 final completed trials and 22 retained full responses, including the truncated Opus attempt.

Token-price estimates for retained responses are $9.09 for Astra, $4.79 for Opus, and $14.83 for Fable ($28.71 total). These include the retained truncated attempt but exclude the lost Astra response and small endpoint smoke tests; they are not billing statements.

Prompts and full transcripts

Each "Prompt" link contains the complete original user prompt, including all weights. Each model link contains the entire API request (including the system instruction and settings) and the entire returned response JSON. No returned text is removed; available reasoning fields, stop reasons, and usage are included. Authentication headers are excluded. Reasoning that the provider did not return cannot be included. Links assume publication of this repository's master branch at the GitHub repository.

Task	Input	Astra	Opus 5.5	Fable 5.1
Pac-Man	Prompt	Transcript	Transcript	Transcript
Breakout	Prompt	Transcript	Transcript	Transcript
Pong	Prompt	Transcript	Transcript	Transcript
Freeway	Prompt	Transcript	Transcript	Transcript
Circle	Prompt	Transcript	Transcript	Transcript
Square	Prompt	Transcript	Transcript	Transcript
Triangle	Prompt	Transcript	Transcript	Transcript

Earlier truncated Opus Breakout response: full transcript.

Scoring and provenance: final per-case scores, all recorded attempts, manual adjudication, plans and incident records, runner and scorer.

Assistant assessment

Success on these seven published inputs is not exclusive to Astra: Opus and Fable also identify all game and shape categories. This supports cross-model replicability of these demonstrations, not broad weight-space understanding. The repeated circle error illustrates the distinction between recognizing a category and accurately describing the complete network's output.

These results do not identify how any model acquired the ability. They cannot establish or rule out task-specific training, reinforcement learning, memorization, or transfer from broader reasoning and interpretability knowledge. Only seven public checkpoints were tested, with one final completed response per model and checkpoint. No fresh-network, description-only, or shuffled-weight control was run. The study therefore provides limited evidence about out-of-distribution generalization or the original training-history hypothesis.